

Disproof of TLMC Conjecture 00000000122

Prime values of q -Catalan numbers

TLMC submission

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Conjecture 1 (TLMC 00000000122). *For every $n \geq 2$ there exist infinitely many primes q such that the n -th q -Catalan number $C_n(q)$ is prime.*

Theorem 2 (Verdict). *Conjecture 2 is **false**. The instance $n = 2$ admits at most one prime value of q under both standard readings of the q -Catalan numbers.*

1 The attack: $n = 2$

1.1 Standard (Gaussian-binomial) reading

Under the standard definition

$$C_n(q) = \frac{\begin{bmatrix} 2n \\ n \end{bmatrix}_q}{\begin{bmatrix} n+1 \\ n \end{bmatrix}_q}, \quad \begin{bmatrix} a \\ b \end{bmatrix}_q = \prod_{i=1}^b \frac{1 - q^{a-b+i}}{1 - q^i},$$

one computes

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = \frac{(1 - q^3)(1 - q^4)}{(1 - q)(1 - q^2)} = \frac{[3]_q [4]_q}{[1]_q [2]_q} = (1 + q^2)(1 + q + q^2),$$

so that, since $[3]_q = 1 + q + q^2$,

$$C_2(q) = \frac{(1 + q^2)(1 + q + q^2)}{1 + q + q^2} = q^2 + 1. \tag{1}$$

Lemma 3. *If q is an odd integer with $q \geq 3$, then $q^2 + 1$ is not prime.*

Proof. q odd implies q^2 odd, hence $q^2 + 1$ even, i.e. $2 \mid q^2 + 1$; concretely $q = 2j + 1$ gives $q^2 + 1 = 2(2j^2 + 2j + 1)$. Moreover $q \geq 3$ gives $q^2 + 1 \geq 10 > 2$. A number divisible by 2 and strictly larger than 2 is not prime. $\square$

Every odd prime q satisfies the hypotheses of Lemma 3. Hence the only prime q for which $C_2(q)$ can be prime is $q = 2$; and indeed $C_2(2) = 5$ is prime. The $n = 2$ instance therefore has exactly one witness — not infinitely many.

1.2 Carlitz reading

Under the Carlitz recurrence $C_0(q) = 1$, $C_{n+1}(q) = \sum_{k=0}^n q^k C_k(q) C_{n-k}(q)$, one gets $C_1(q) = 1$ and

$$C_2(q) = C_0 C_1 + q C_1 C_0 = 1 + q.$$

The identical argument applies: for odd prime q , $q + 1$ is even and ≥ 4 , hence composite, and $C_2(2) = 3$ is prime. Exactly one witness.

Theorem 4 (Exact classification for $n = 2$). *Let q be prime. Under the standard reading, $C_2(q)$ is prime if and only if $q = 2$ (with $C_2(2) = 5$). Under the Carlitz reading, $C_2(q)$ is prime if and only if $q = 2$ (with $C_2(2) = 3$).*

2 Boundary of the disproof

Remark 5. *The counterexample is the single instance $n = 2$. Nothing here excludes the conjecture for any fixed $n \geq 3$: the falsity is a failure of the universal quantifier over n , not of the infinitude phenomenon for each individual degree. The surviving witness $q = 2$ is genuine in both readings, so the failure is exactly “one prime witness instead of infinitely many”.*

3 Lean certificate

The attack is formalized in core Lean 4 (no Mathlib), toolchain `leanprover/lean4:v4.33.1`, in the file `lean4/Main.lean`. Headline statements:

- `tlmc122_disproof_standard`: for every prime q , $\text{IsPrime}(C_2(q)) \rightarrow q = 2$ with $C_2(q) = q * q + 1$;
- `tlmc122_disproof_carlitz`: the same for $C_2(q) = q + 1$;
- `C2_prime_iff_eq_two`, `C2Carlitz_prime_iff_eq_two`: the exact classification of Theorem 4, including the primality of 5 and 3.

The development uses a self-contained primality predicate and a decomposition-based parity API, deliberately avoiding `omega`, `simp` and the core `mod/div` lemmas (which carry `propext` or `Quot.sound` in this toolchain). The audit `lean4/Check.lean` runs `#print axioms` on all 21 theorems; every one reports *does not depend on any axioms*. There is no `sorry`.

4 Reproduction

`reproduce.py` recomputes everything independently with exact integer polynomial arithmetic:

1. the Gaussian binomial $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q$ by the product formula with exact division, confirming $C_2(q) = q^2 + 1$;
2. the Carlitz polynomials through C_3 , confirming $C_2(q) = 1 + q$;
3. an enumeration of all primes $q < 500$: in both readings the only prime q with $C_2(q)$ prime is $q = 2$.

The script asserts each of these facts and exits non-zero on any mismatch.